

Project Report

Streamlining IT Procurement: Automating Standard Laptop Orders With Flow Designer

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1. INTRODUCTION

1.1 Project Overview

In most organisations, employees request standard laptops through emails, spreadsheets or phone calls. Each request then has to be checked, approved, ordered and delivered by hand, which is slow, hard to track and prone to errors. This project automates that process on the ServiceNow platform using **Flow Designer**, its low-code workflow tool.

An employee raises a request for a standard laptop from a Service Catalog item. Flow Designer then runs the whole lifecycle: it validates the request, routes it for manager approval, creates fulfilment tasks for the procurement team, updates the request status and sends notifications to the requester at each stage. No scripting-heavy development is needed, so the workflow is easy to maintain and modify.

1.2 Purpose

- Reduce the time taken to process standard laptop requests by replacing manual steps with an automated flow.
- Give employees a single, simple catalog form to request a laptop.
- Enforce a consistent approval and procurement process with a clear audit trail.
- Give requesters and IT staff real-time visibility into the status of every request.
- Show how low-code automation in Flow Designer can streamline routine IT procurement.

2. IDEATION PHASE

2.1 Problem Statement

Employees who need a standard laptop have no single, structured way to request one. Requests arrive by email or phone, approvals are chased manually, and the procurement team has no central view of what is pending. This causes delays, duplicate or missing requests, and frustration for employees, managers and IT staff.

I am	I am trying to	But	Because	Which makes me feel
An employee needing a laptop	Get a standard laptop quickly	I do not know where to request it or how far it has got	The process is manual and scattered across emails and calls	Frustrated and uncertain
A manager	Approve laptop requests	Requests are lost in my inbox	There is no approval workflow	Overloaded
An IT procurement officer	Fulfil requests efficiently	Details are incomplete and tracking is manual	There is no standard request form or central queue	Stressed and unproductive

2.2 Empathy Map Canvas

Says	Thinks
“How long will my laptop take?” “Who do I ask for approval?” “Can someone update me on my request?”	Why is such a simple request so slow? Will my request be forgotten? There should be an easier way.
Does	Feels
Sends emails and follow-ups Calls the IT help desk Waits for approvals and delivery	Frustrated by delays Anxious about project deadlines Relieved when requests are visible and tracked
Pains	Gains
Long waiting times, no status visibility, repeated follow-ups, manual data entry errors	Fast, standard request form, automatic approvals, status notifications, less paperwork for IT

2.3 Brainstorming

The team listed ideas to address the problem and then grouped them by impact and effort.

Idea	Impact	Effort	Decision
Service Catalog item for standard laptop requests	High	Low	Selected
Automated manager approval using Flow Designer	High	Low	Selected
Auto-creation of catalog tasks for the procurement team	High	Medium	Selected
Email notifications at each request stage	Medium	Low	Selected
Dashboard for request tracking and reporting	Medium	Medium	Future scope
Integration with external vendor ordering systems	High	High	Future scope

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	Employee action	Touchpoint	Experience / System response
Awareness	Needs a new standard laptop	ServiceNow portal	Finds the laptop request item in the catalog
Request	Fills in the catalog form and submits	Service Catalog form	Request record (RITM) is created; confirmation is sent
Approval	Waits for manager decision	Email / approval list	Flow routes approval to the manager automatically
Fulfilment	Tracks progress	Request status page	Catalog task is assigned to the procurement team
Delivery	Receives laptop	IT / procurement team	Task is closed and the request is marked complete
Feedback	Confirms satisfaction	Notification email	Requester is notified; process can be improved

3.2 Solution Requirements

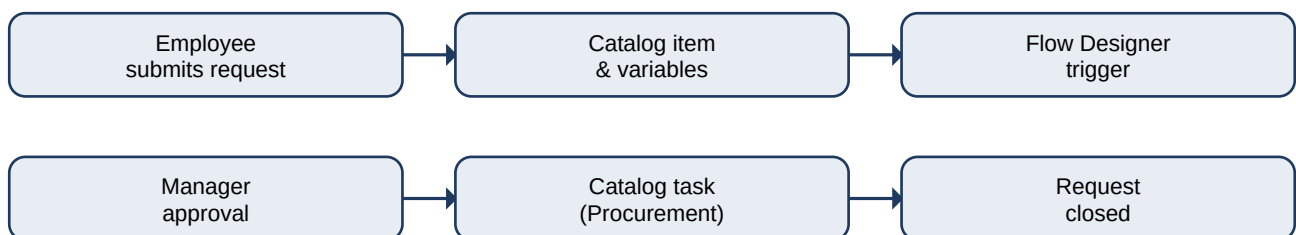
Functional requirements

ID	Requirement	Description
FR-1	Catalog request	User can submit a standard laptop request with required details through a Service Catalog item.
FR-2	Automated approval	The flow sends the request to the requester's manager for approval.
FR-3	Task creation	On approval, a fulfilment task is created and assigned to the procurement group.
FR-4	Status update	Request state is updated automatically as approval and fulfilment progress.
FR-5	Notifications	The requester is notified on submission, approval or rejection, and completion.
FR-6	Rejection handling	A rejected request is closed with the outcome communicated to the requester.

Non-functional requirements

Type	Requirement
Usability	Simple catalog form that needs no training.
Security	Role-based access; only authorised users can approve or fulfil requests.
Reliability	Flow executes consistently for every request, with execution history for troubleshooting.
Maintainability	Low-code flow that administrators can modify without heavy scripting.
Traceability	Complete audit trail of approvals, tasks and state changes.

3.3 Data Flow Diagram



Data entered by the employee is stored in the request (RITM) record. The flow reads this data, creates an approval record for the manager, and on approval generates a catalog task. Status changes and notifications are written back to the request so that the employee can see progress at any time.

3.4 Technology Stack

Component	Technology
Platform	ServiceNow (Personal Developer Instance)
Workflow automation	Flow Designer (triggers, actions, approvals, flow logic)

Component	Technology
Request interface	Service Catalog item with variables / Service Portal
Data storage	ServiceNow tables (Requested Item, Approval, Catalog Task)
Notifications	ServiceNow email notifications
Access control	Users, groups and roles

4. PROJECT DESIGN

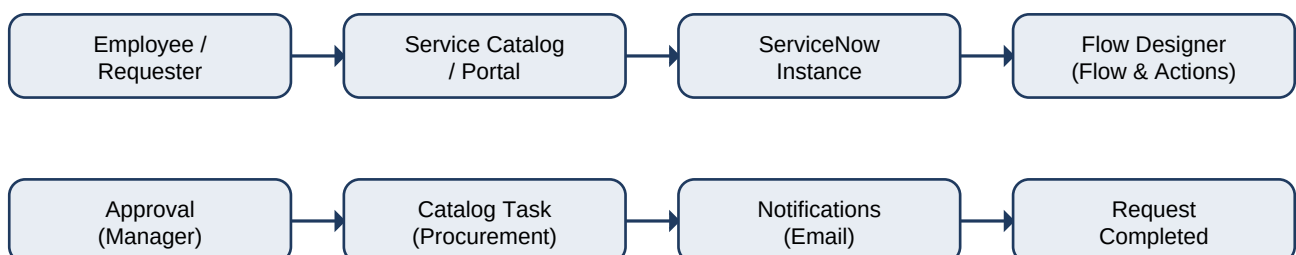
4.1 Problem Solution Fit

Problem	Solution fit
Requests are made through scattered channels	One standard catalog item for every laptop request
Approvals are chased manually	Flow sends approval automatically and records the decision
No visibility of request status	Automatic state updates and notifications keep everyone informed
Procurement team lacks a central queue	Approved requests generate tasks assigned to the procurement group
Processes differ from request to request	The same flow runs for every request, ensuring consistency

4.2 Proposed Solution

Aspect	Description
Novelty	Applies low-code Flow Designer to the everyday problem of standard laptop procurement.
Feasibility	Uses only built-in ServiceNow capabilities; no external tools are required.
Business impact	Faster fulfilment, lower manual effort and better tracking of IT hardware requests.
Customer satisfaction	Employees get a simple request form, clear updates and quicker delivery.
Scalability	The flow can be extended to other standard assets such as monitors, phones or accessories.

4.3 Solution Architecture



The employee interacts with the catalog item, which creates a request record on the ServiceNow instance. Flow Designer is triggered by the request and orchestrates approval, task creation, status

updates and notifications.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Sprint	Activity	Planned outcome
Sprint 1	Requirement study, problem definition, instance setup	Problem statement, empathy map, users and groups created
Sprint 2	Design and build the catalog item and variables	Standard Laptop Request catalog item available
Sprint 3	Build the flow: trigger, approval, task creation, status updates	Working end-to-end flow in Flow Designer
Sprint 4	Notifications, testing, UAT and documentation	Tested solution, UAT report and final project report

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

The solution was tested against the scenarios below. Detailed UAT results are given in the separate UAT report.

ID	Scenario	Expected result
TC-1	Submit a laptop request with all required fields	Request is created and confirmation is sent to the requester
TC-2	Submit a request with a mandatory field missing	Form prevents submission and highlights the missing field
TC-3	Manager approves the request	Flow continues and a catalog task is created for procurement
TC-4	Manager rejects the request	Request is closed as rejected and the requester is notified
TC-5	Procurement team completes the task	Request is marked complete and the requester is notified
TC-6	Check flow execution history	Each step of the flow is logged and can be reviewed
TC-7	Multiple requests submitted together	Each request runs its own flow instance without interference

7. RESULTS

7.1 Output Screenshots

The screenshots below show the working solution. Insert the corresponding screenshots in the marked spaces.

[Insert screenshot here]

Figure 1: Standard Laptop Request catalog item

[Insert screenshot here]

Figure 2: Flow in Flow Designer (trigger, approval, task and notification actions)

[Insert screenshot here]

Figure 3: Manager approval record

[Insert screenshot here]

Figure 4: Catalog task created for the procurement team and request completion

8. ADVANTAGES & DISADVANTAGES

Advantages

- Faster processing of standard laptop requests with minimal manual effort.
- Consistent approval and fulfilment process for every request.
- Clear status visibility and automatic notifications for requesters.
- Complete audit trail of approvals, tasks and state changes.

- Low-code design that is easy to modify and extend.
- Uses native ServiceNow features, so no extra software or integration is required.

Disadvantages

- Depends on the ServiceNow platform and appropriate licensing in a production environment.
- Covers only standard laptop orders; non-standard or custom hardware needs a separate process.
- Does not yet connect to external vendor or inventory systems.
- Requires up-to-date user, manager and group data for approvals to route correctly.

9. CONCLUSION

This project shows that a routine IT procurement task can be automated effectively with ServiceNow Flow Designer. By combining a Service Catalog item with an automated flow for approval, task creation and notifications, the solution replaces scattered manual steps with one consistent, traceable process. Employees benefit from a simple request experience, managers from structured approvals, and the procurement team from a clear work queue. The same approach can be reused for other standard IT requests.

10. FUTURE SCOPE

- Extend the flow to other standard assets such as monitors, mobile phones and accessories.
- Integrate with vendor systems for automatic purchase orders and delivery tracking.
- Add inventory checks so that available stock is allocated before a new purchase is raised.
- Build dashboards and reports for request volume, approval time and fulfilment time.
- Add multi-level approvals based on cost or department, and SLA tracking with escalations.
- Provide a mobile-friendly experience for raising and approving requests.

11. APPENDIX

Source Code

The project is built using ServiceNow configuration (catalog item, flow and notifications) rather than custom application code. The configuration and project files are available in the GitHub repository below.

Dataset Link

Not applicable. The project uses records created within the ServiceNow instance and does not use an external dataset.

GitHub & Project Demo Link

GitHub	https://github.com/karthikar-12/Streamlining-IT-Procurement-Automating-Standard-Laptop-Orders-With-Flow-Designer
Demo Link	https://drive.google.com/file/d/1JE4BSQaxmTknBbU6Ixr0fwZvhYc4nJJ/view?usp=drivesdk

Document Link	https://drive.google.com/file/d/1fd2vbx21ivYQ8ffxW2LJMr7l-ucZmzf5/view?usp=drivesdk
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